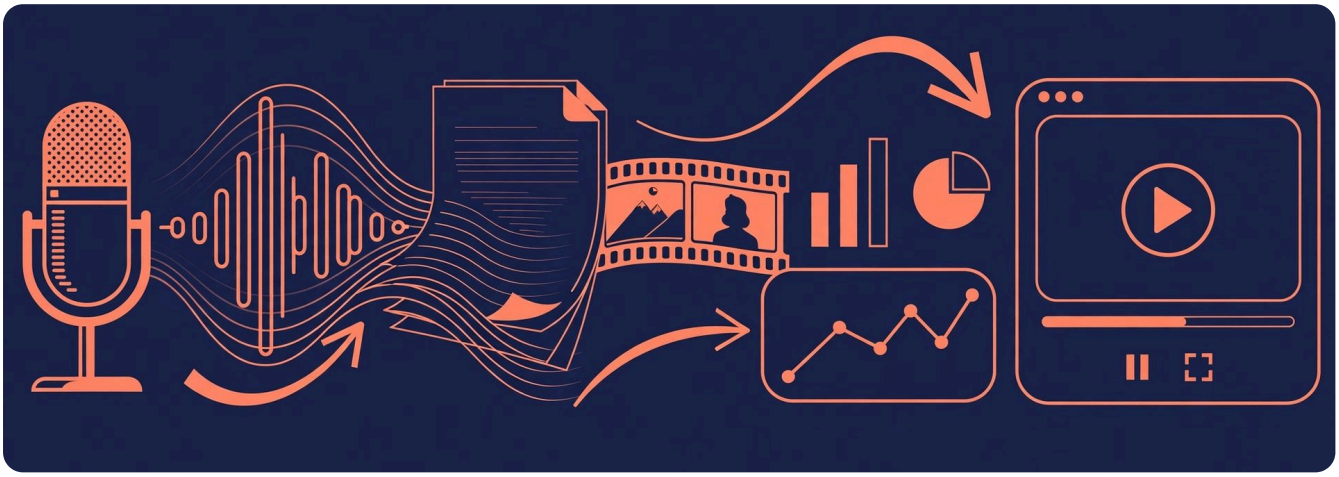
BUILD PLAN - OCTOBER 2026

AI Video Studio

Our own version of Slot Studio: topic and voiceover in, a finished video, Shorts, thumbnails and publish-ready metadata out - as a web app that works on Windows and Mac.



> **Verdict: Yes, we can build this.** We already have the hardest basics in-house - transcription and voice cleanup with faster-whisper, ffmpeg video assembly from our reels pipeline, and LLM scripting. The new work is the orchestrator that chains them, plus a motion-graphics template library in Remotion. That is a product build, not a research project.

What this plan covers

- What Slot Studio actually is, and where we deliberately differ
- The 9-stage pipeline, module by module, with inputs and outputs
- Architecture, tech stack and hosting
- Phased roadmap: validation spike, MVP, sellable v1, and v2 polish
- Running costs, risks, and the legal line on clips
- The one decision needed to start

Timelines in this plan are estimates for one developer building full-time with AI assistance, based on reusing our existing Whisper, ffmpeg and scripting code. They are not guarantees - the validation spike in Phase 0 exists to firm them up.

Slot Studio is a Claude pack, not a platform - we build a platform

Slot Studio (slotstudio.app) is not standalone software. It is a folder added to the Claude desktop app as a Project. It needs a Mac with an M1 chip or newer and the user's own paid Claude plan. Input is a topic plus a voiceover; output is a rendered 4K video with thumbnails, Shorts and metadata.

Point	Slot Studio	Our version
Form	Claude Project folder, runs inside Claude desktop	Web app - browser in, video out, any OS
Platform	Mac only (M1+)	Windows, Mac, Linux - rendering is server-side
AI cost	User brings a Claude plan; Max tier recommended by them	We control the model per stage - cheap models for cleanup, strong models for script
Price seen	Rs 24,999 one-time + GST until 23 Oct, then Rs 34,999, plus the Claude subscription	Our pricing is ours to set - see Costs page
Visuals	Finds YouTube clips and photos	User footage + licensed stock first; YouTube clips only as an opt-in with a risk warning
Render	Local, about 95 minutes quoted for a 6-minute video on an M5	Server render queue; 1080p default, 4K as a slower tier

Where we win, where we do not

Advantages

- No Mac or Claude subscription needed by the user
- Works for our existing Pixalified clients on Windows
- We own the pipeline - templates, voices and styles compound as assets
- Can plug into our current reels workflow for Shorts/Reels

Honest gaps

- Their template polish is the product. Matching it takes the most time.
- 4K rendering is slow and expensive anywhere - theirs is slow too.
- A folder pack has zero hosting cost. A web app has ongoing infra.

One job, nine stages: topic and voice in, publish pack out

1 Topic > 2 Research > 3 Script > 4 Voice > 5 Visual plan > 6 Graphics > 7 Edit + sound > 8 Render > 9 Publish pack

Stage	What happens	Output
1. Intake	User enters topic, target length, language, style preset and uploads a voiceover - or picks AI voice.	Job with settings
2. Research	Web search and source fetch; facts extracted with sources kept attached to each claim.	Research brief with sources
3. Script	LLM writes hook, structure and narration in the channel's voice. User can edit before anything renders.	Editable script, scene list
4. Voice	User voice cleaned: breaths, filler words and long silences removed. Or TTS generated per scene.	Clean timed audio + word timings
5. Visual plan	Each scene matched to footage, a stock clip, a photo or a graphic. User approves swaps.	Storyboard
6. Graphics	Remotion templates render charts, comparisons, tier lists, quotes, maps and lower-thirds from scene data.	Graphic clips per scene
7. Edit + sound	Assembly, animated captions from word timings, background music ducked under voice, sound effects on transitions.	Preview video (low-res, fast)
8. Render	Full render after user approves the preview. 1080p standard; 4K optional.	Master MP4
9. Publish pack	Auto-cut 2-3 vertical Shorts/Reels, 3 thumbnail options, title, description, chapters and hashtags.	Downloadable pack (.zip)

The approval gates are the design. Script approval, storyboard approval and preview approval keep the expensive steps (render, TTS credits) from being wasted, and keep a human in control of facts and visuals. Nothing publishes automatically.

Architecture: a thin web app over a render pipeline

Layer	Choice	Why
Frontend	Next.js (React) + Tailwind	Fast to build, our usual stack, good file-upload and preview UX
API / orchestrator	FastAPI (Python) job service	Our Whisper and ffmpeg code is already Python; one orchestrator runs the 9 stages as resumable steps
Job queue	Redis + Celery (or RQ)	Renders are long. Jobs must survive restarts, retry per stage, and report progress
Speech	faster-whisper (self-hosted) + ffmpeg silencedetect	Already working in-house; word timings drive captions and filler removal
LLM	Claude / GPT via API, model set per stage	Strong model for script, cheap model for cleanup and metadata
TTS (optional)	ElevenLabs or OpenAI TTS, behind a provider switch	Only for users without a voiceover; cost passed through as credits
Graphics + edit	Remotion (React) templates, rendered headless	Charts and captions as code - reusable, brandable per channel
Final assembly	ffmpeg	Already our reels pipeline: loudness, ducking, burn-in fallback, exports
Storage	S3 / Cloudflare R2 + Postgres	Assets, renders and job state; signed download links
Stock media	Pexels / Pixabay APIs (licensed)	Free licensed footage and photos with attribution tracked per asset

Key design rules

- **Every stage is resumable.** A failed render retries that stage, not the whole job.
- **Preview before master.** A 540p preview renders in minutes; the 1080p/4K master only runs after approval.
- **Templates are data-driven.** A chart scene is JSON (title, values, labels) fed to one Remotion component - new chart types are new components, not new pipelines.
- **Provenance is stored.** Every fact in the script and every clip keeps its source, shown to the user at approval.

Roadmap: prove it on one video, then build the product

Phase	Estimate	Scope and exit test
0. Validation spike	3-4 days	No UI. Script one real topic end-to-end with CLI tools: research, script, Whisper cleanup, 3 Remotion graphics, ffmpeg assembly. **Exit:** one 3-5 minute video we would actually post. If quality fails here, stop cheap.
1. MVP	3-4 weeks	Web app with login, job creation, script and storyboard approval, voice upload + cleanup, 6 graphic templates, animated captions, 1080p render, 1 auto Short, title/description. Single user style preset. **Exit:** 5 videos produced through the UI without developer help.
2. Sellable v1	+4-6 weeks	Payments and credits, TTS option, thumbnail generator (3 options), 2-3 Shorts per video, 12+ templates, per-channel brand kits (fonts, colours, logo), music library, job history and re-renders. **Exit:** a paying creator completes a video start to finish.
3. v2 polish	Ongoing	4K tier, Hindi/Hinglish script and caption tuning, team accounts, template marketplace, analytics on Shorts performance, batch topics.

Template library - the real asset

MVP ships six: animated bar chart, line chart, comparison (A vs B), tier list, big-quote card, and stat counter. V1 adds pie/donut, timeline, map highlight, screenshot callout, list reveal and chapter title. Each template is roughly half a day to a day once the Remotion harness exists - the harness is the hard part, templates are routine after it.

What we reuse from existing work

faster-whisper setup | ffmpeg reels pipeline | LLM script prompts | caption styling | Vercel + VPS deploy patterns

Reuse is why the MVP is weeks, not months. What does not exist yet: the Remotion harness, the storyboard UI, and the job queue - those three are most of Phase 1.

Costs are dominated by renders and AI calls, not hosting

Figures below are indicative planning ranges, not quotes. They assume low early volume (tens of videos a month) and should be re-checked against live API pricing before we set a selling price.

Item	Indicative cost	Notes
Hosting (web + API)	Low - existing VPS / Vercel tier	Render workers are the scaling cost, not the website
Render worker	One mid-size VPS to start	Scale by adding workers; queue absorbs spikes
Storage + delivery	Low per video	R2 has no egress fees; masters can expire to cold storage
LLM per video	Low - a few rupees to tens of rupees	Research + script + metadata; cheap models for cleanup steps
TTS per video (optional)	The biggest per-video API cost	Only when the user has no voiceover; charge as credits
Stock media	Free tier to start	Pexels/Pixabay APIs; paid stock later for quality

Suggested pricing shape (to decide, not decided)

- **For our own channels first:** zero marginal pricing questions - build it, use it for Pixalified content, prove quality.
- **If sold:** per-video credits or a monthly plan with included renders. A one-time price like Slot Studio's leaves us carrying render costs forever, so subscription or credits fit our cost shape better.

Risks: the real ones, all manageable

Risk	Mitigation
Clip copyright / YouTube terms	Default to user footage and licensed stock. Downloading YouTube clips for reuse can breach YouTube's terms and the clip owner's copyright. If offered at all, it is an explicit opt-in with the risk stated - never the default path.
Quality gap vs their polish	Phase 0 spike judges one real video before we commit. Templates improve per video; the library compounds.
Render time and cost	540p previews, 1080p default, 4K as a paid slower tier. Queue with per-stage retries so nothing restarts from zero.
Factual errors in scripts	Sources stay attached to claims; script approval gate is mandatory, not skippable.

Recommendation: start with the 3-4 day spike

Do not approve the full build yet. Approve one real video, end to end, with no UI.

The spike, concretely

- Pick one real topic from our own content queue - a video we would genuinely post.
- Run research and script generation; you approve or edit the script.
- Record or supply a rough voiceover; we clean it with the existing Whisper setup.
- Build three Remotion graphics for its scenes - chart, comparison, quote card.
- Assemble with ffmpeg: captions, music ducking, one vertical Short cut.
- Judge the output together against a simple bar: would we post this, and what exactly is missing?

> **Decision after the spike:** if the video is postable, Phase 1 (MVP, 3-4 weeks) starts with known quality and firmer timelines. If it is not, we have spent days, not weeks, and we know which stage failed - graphics, script or voice - before spending more.

Open questions for you

- First user: ourselves (Pixalified content) or an external creator? This changes MVP scope - external means accounts and payments move earlier.
- Language priority: English, Hindi/Hinglish, or both from the start?
- Voice: your recorded voiceover as the default, or AI voice?

Source for the Slot Studio comparison: slotstudio.app, checked 17 October 2026 - form (Claude Project folder), Mac M1+ requirement, bring-your-own Claude plan, pricing shown on their page, and the render-time example quoted on their site. Their page and terms can change; re-check before any public comparison claim.